General Mathematics — Exercise Sheet 34

Second-order linear equations

Variant: Statements

Exercise 1 (A parameter and exceptional initial data). For every $a \in \mathbb{R}$, solve

$$y'' + 2ay' + (a^2 - 1)y = 0, \qquad y(0) = 3, \quad y'(0) = -2.$$

Determine exactly which parameters give a bounded solution on $[0, \infty)$, and which give a solution tending to zero. Include any parameter at which the coefficient of a growing exponential vanishes. Use Lecture 34's complete characteristic family and Lecture 6's exponential limits.

Exercise 2 (Forcing at a double root). Solve

$$y'' + 2y' + y = 6e^{-t}, \qquad y(0) = 2, \quad y'(0) = -1.$$

Use Lecture 34's shifted-expression identity and its double-root forcing rule. Verify both the equation and the two initial values.

Exercise 3 (A third-order equation through the same factorisation method). Prove that every C^3 solution on an open interval containing zero of

$$y''' - y'' - y' + y = 0$$

has the form $(c_1 + c_2 t)e^t + c_3 e^{-t}$. Then find the unique solution with $y(0) = 1$, $y'(0) = 0$ and $y''(0) = 2$. Use Lecture 34's repeated-root second-order family, Lecture 33's integrating factor and Lecture 8's product rule. No general third-order existence theorem is given.

Exercise 4 (Critical damping and a future crossing). Let $\gamma > 0$ and $v_0 \in \mathbb{R}$. The critically damped motion satisfies

$$x'' + 2\gamma x' + \gamma^2 x = 0, \qquad x(0) = 1, \quad x'(0) = v_0.$$

Determine exactly when x crosses zero at a positive time. Find that time when it exists, and determine the sign on either side. Also find the set of initial velocities for which $x(t) > 0$ for every $t \geq 0$. Use Lecture 34's repeated-root family and positivity of the real exponential.

Exercise 5 (A certified complex exponential approximation). Use Lecture 34's complex factorial series to compute $S_8 = \sum_{n=0}^{8} (1+\mathrm{i})^n/n!$ exactly. Prove $|e^{1+\mathrm{i}} - S_8| \le \sqrt{2}/18144 < 1/10000$. Deduce absolute error bounds for both $e\cos 1$ and $e\sin 1$ from the real and imaginary parts of S_8. Use Lecture 29's ratio-tail argument, Lecture 15's complex modulus and Lecture 31's component convergence. Angles are in radians.